
Opening the Black Box:

Localizing semantic inconsistency in NLI models with Deep k -Nearest Neighbors

PRESENTED BY

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What we will cover

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|----|----------------------|---|
| 01 | The NLI task | Premise, hypothesis, and the three logical relations |
| 02 | The problem | Why strong models stay fragile, as the literature shows |
| 03 | The gap and our idea | Opening the black box with Deep k -Nearest Neighbors |
| 04 | Methodology | Locating <i>where</i> inconsistency begins, the core of this talk |
| 05 | Summary | The gap, the questions, and our angle of attack |

Inference as a test of understanding

Natural Language Inference is a core task of Natural Language Understanding — whether a model truly grasps meaning, not just surface patterns in the text.



If a model can reliably judge the relation between two sentences, we credit it with understanding the language, at least in part.

Premise and hypothesis

Premise

The given fact, the original sentence we take as true.

Hypothesis

The sentence we test *against* the premise. The model returns their relation.

PREMISE

Danny is eating ice cream.

HYPOTHESIS

Danny is eating dessert.

RELATION

Entailment

Three possible relations

Entailment

The hypothesis follows from the premise.

Premise: The boy is sleeping.
Hypothesis: The boy is resting.

Contradiction

The hypothesis conflicts with the premise.

Premise: The boy is sleeping.
Hypothesis: The boy is running.

Neutral

Neither follows nor conflicts.

Premise: The boy is sleeping.
Hypothesis: The boy is dreaming.

Neutral reflects genuine logical independence, *not* the model being uncertain.

Paraphrase: same meaning, new words

A paraphrase rewords a sentence while preserving its meaning. The anchor paper verifies this with symmetric entailment — each sentence must entail the other.

ORIGINAL HYPOTHESIS

Danny is eating dessert.



SYMMETRIC
ENTAILMENT

PARAPHRASE

Danny is having a sweet dish.

Same logic, different surface form. A robust model should return the same relation for both.

PART 02

The problem

High benchmark scores, fragile predictions — what the literature already knows.

Semantic Sensitivities and Inconsistent Predictions

Measuring the Fragility of NLI Models

Arakelyan et al., 2024 • EACL 2024

THE CENTRAL CLAIM

Even strong Transformer-based NLI models change their predictions under small, meaning-preserving rewordings.

A diagnosis, not a fix

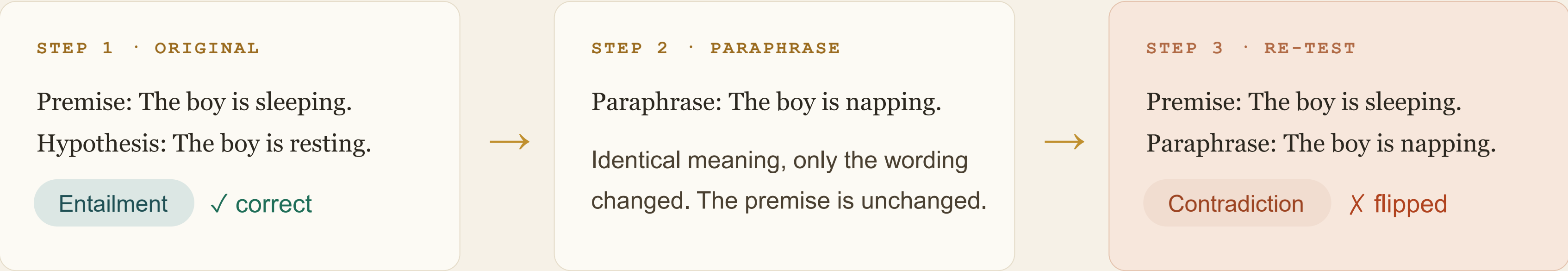
Its contribution is a framework for measuring semantic sensitivity, not a new model.

Why it matters

It sharpens a central NLP debate, do language models understand meaning or just match patterns.

Semantic inconsistency, illustrated

Arakelyan et al., 2024



Same meaning. Opposite prediction. This is semantic inconsistency.

How fragility is measured

Arakelyan et al., 2024

- 1 Keep only examples the model already gets right.
- 2 Generate about five meaning-preserving paraphrases per hypothesis.
- 3 Re-test, and count how often the prediction changes.

Relaxed Fooling Rate

Any change in the prediction after paraphrasing.

Strict Fooling Rate

A full flip, Entailment becomes Contradiction.

What they found

Arakelyan et al., 2024

12.9%

average Relaxed
Fooling Rate, in-domain

23.7%

average Relaxed Fooling
Rate, out-of-domain

- Smaller models are more fragile, distilled models lose the most stability.
- Even very large models still suffer from semantic inconsistency.
- The effect is worse in the zero-shot setting.

The same verdict, again and again

Right for the Wrong Reasons

McCoy et al., 2019

Models lean on syntactic heuristics like lexical overlap, accuracy collapses when the shortcut fails.

Breaking NLI Systems

Glockner et al., 2018

High SNLI scores, yet failure on simple lexical inference like synonyms and antonyms.

Stress Test Evaluation

Naik et al., 2018

Accuracy drops under negation, antonymy, word overlap, and noisy input.

Probing with Semantic Fragments

Richardson et al., 2020

Struggle on basic logic, negation, quantifiers, comparatives, and counting.

High benchmark accuracy is not real language understanding.

The field kept measuring the output

How Often Are Errors Due to Paraphrastic Variability

Srikanth et al., 2024

Introduces Paraphrastic Consistency , the chance a model stays consistent on a pair.

MERGE

Zgreabănă et al., 2025

Proposes Pattern Accuracy , consistency over a whole set of paraphrases.

Transitive Self-Consistency

Wu & Last, 2025

Measures consistency across reasoning chains, without gold labels.

Valuable metrics, but all of them score the final output. None ask *where* inside the model it breaks.

PART 03

The gap and our idea

Adapting Deep k -Nearest Neighbors to ask *where*, inside the model, meaning breaks.

Research Gap

WHAT WE KNOW

The phenomenon is real, documented across dozens of models and datasets.

WHAT IS MISSING

- 1 Work diagnoses the problem, it rarely builds mechanisms that improve consistency.
- 2 It is unclear how to train models to stay consistent under semantic variation.
- 3 No one has opened the black box to ask *where* the inconsistency originates.

Everyone measured the result. No one localized the cause.

Deep k -Nearest Neighbors

Papernot & McDaniel, 2018 *Deep k -Nearest Neighbors: Towards Confident, Interpretable and Robust Deep Learning*

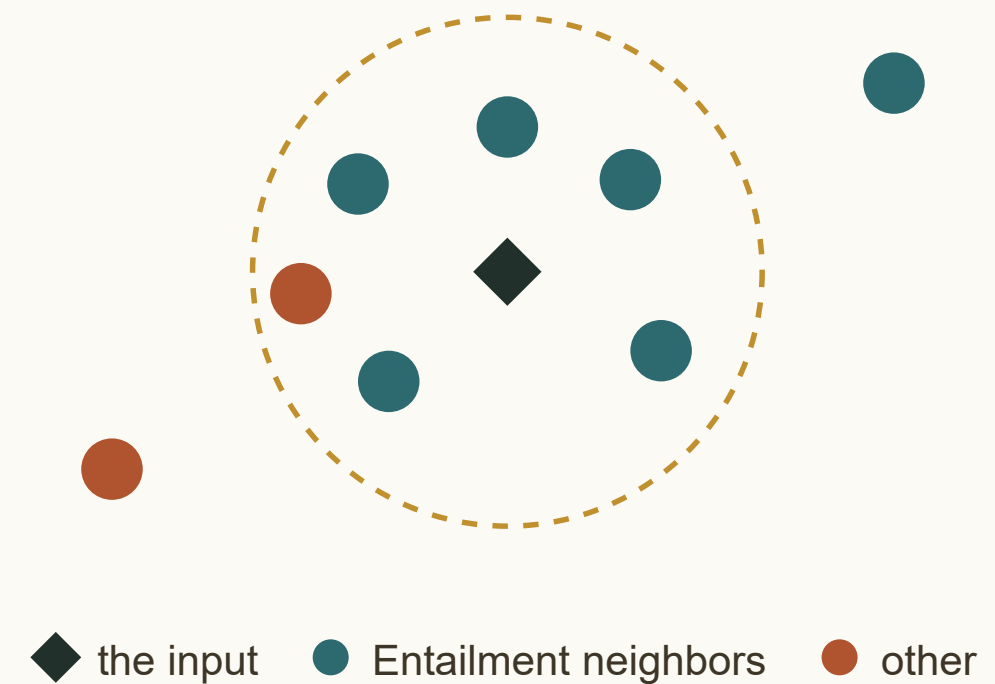
THE USUAL WAY

We read only the model's final answer and trust it — the reasoning inside stays hidden.

DEEP K-NN

At every layer it asks, which training examples does this input most resemble, and what were their labels .

INSIDE ONE LAYER



Most neighbors are Entailment, so this layer supports Entailment. Repeat across every layer and tally the votes.

Credibility: consistent internal support

A Deep k-NN signal · Papernot & McDaniel, 2018

Across all layers, gather each input's K nearest training neighbors . Credibility is the share of those neighbors that support the final label.

Not just what the model decided, but how consistent its thinking was along the way.

12 layers × K=10 = 120 votes

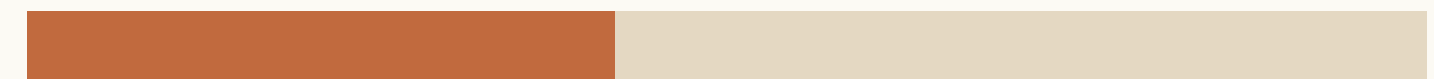
100 of 120 agree

high credibility



50 of 120 agree

low credibility



What we ask

MAIN RESEARCH QUESTION

Does the semantic inconsistency of NLI models arise already in their *internal representations* , not only in the final output?

SECONDARY RESEARCH QUESTIONS

Q1

Is low credibility associated with semantic inconsistency?

Q2

In which layer region is it most pronounced, and what does that imply?

Q3

Can credibility serve as a signal for targeted improvement?

HYPOTHESIS

Consistent pairs show high credibility. Inconsistent pairs show a credibility drop from the hypothesis to its paraphrase.

PART 04

Methodology

Locating *where* meaning breaks — three measurements, two phases, every choice grounded in the literature.

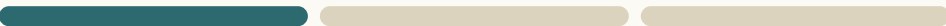
Datasets: SNLI, MNLI, ANLI

SNLI

Bowman et al., 2015

Sentence pairs from image captions, the first large-scale NLI corpus.

EASIEST



MNLI

Williams et al., 2018

Ten genres, from fiction to telephone speech, broad coverage.

MEDIUM



ANLI

Nie et al., 2020

Built adversarially against models, deliberately hard.

HARDEST



The three datasets that recur most across the reviewed work, spanning increasing difficulty. We keep them separate , merging would blur where any improvement came from.

Models under study

RoBERTa-large

DeBERTa-large

BERT-base

BART-large

The models that appear most often across the surveyed work.

Arakelyan et al., 2024 Nie et al., 2020 · Wu & Last, 2025

12 model and dataset pairs (4 models × 3 datasets)

Setting k — a setup decision

TRY SEVERAL

$k = 5$

$k = 10$

$k = 20$

$k = 50$



Pick the k that maximizes the correlation between **low credibility** and **actual inconsistency**.

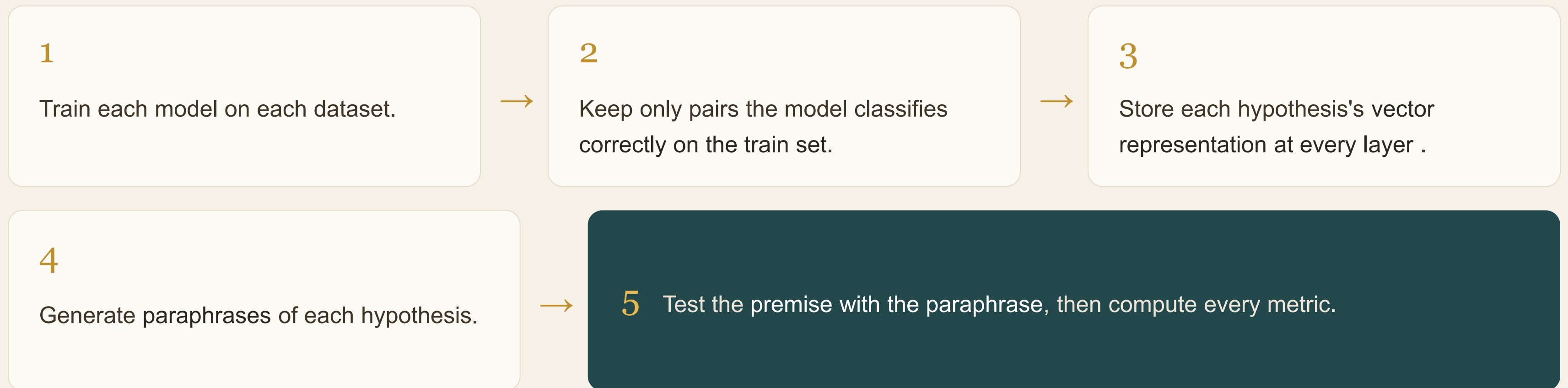
Measured on the validation set, before any experiment is run.

ONE k PER

model $\times$ dataset pair

Different representation spaces
respond differently.

The experimental pipeline



A controlled comparison

✓ Only examples the model already gets right enter, so we measure sensitivity, not ordinary errors.

✓ Each paraphrase preserves the relation , so any change in answer is unjustified.

AFTER RE-TESTING, AN ANSWER IS EITHER

Consistent

Same correct answer, our baseline for comparison.

Inconsistent

The answer changed, the cases we investigate.

Metrics, grounded in the literature

OUTPUT CONSISTENCY

Relaxed & Strict Fooling Rate

Semantic Sensitivities & Inconsistent Predictions Arakelyan, 2024

Paraphrastic Consistency

How Often Are Errors... Srikanth, 2024

Pattern Accuracy

MERGE Zgreabănă, 2025

INTERNAL REPRESENTATION

Layer Distance

vector geometry across layers

Credibility

Deep k-Nearest Neighbors Papernot, 2018

The internal lenses that let us open the black box.

Metric 1 • Consistency

did the answer hold under paraphrasing?

For every hypothesis and paraphrase pair, is the answer the same? Neutral is included, Neutral staying Neutral still counts as consistent.

Relaxed — any change

Entailment → Neutral

Neutral → Contradiction

Every one of these counts as inconsistent.

Strict — full flip only

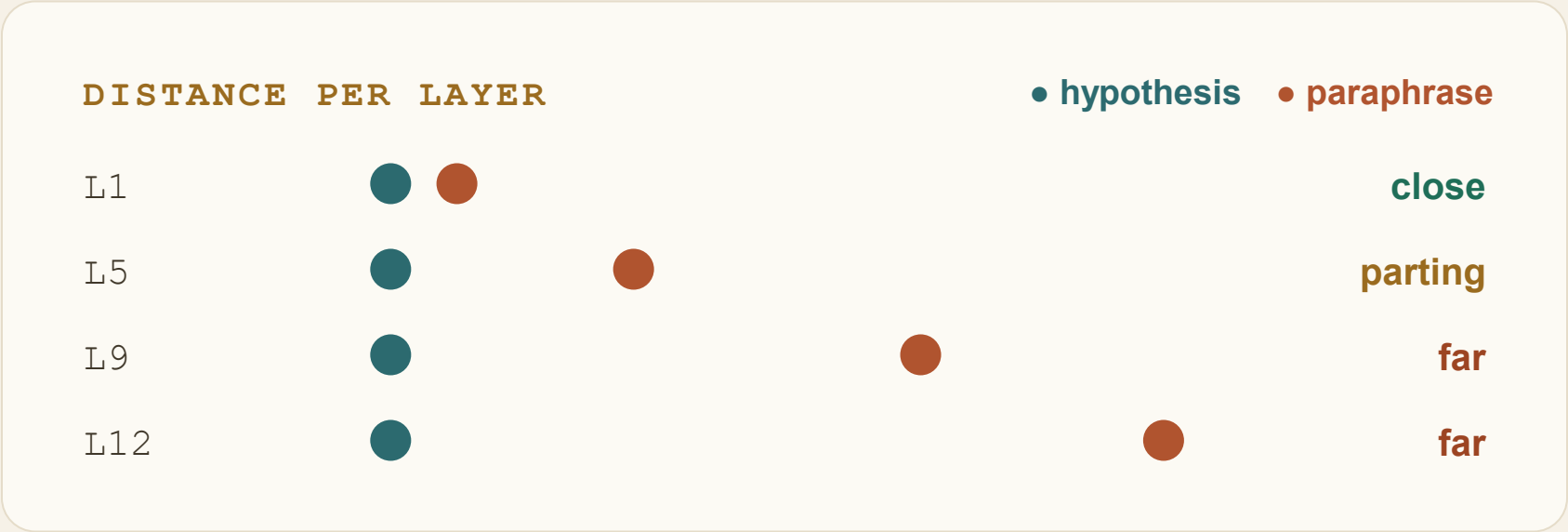
Entailment ↔ Contradiction

Only a complete reversal between Entailment and Contradiction.

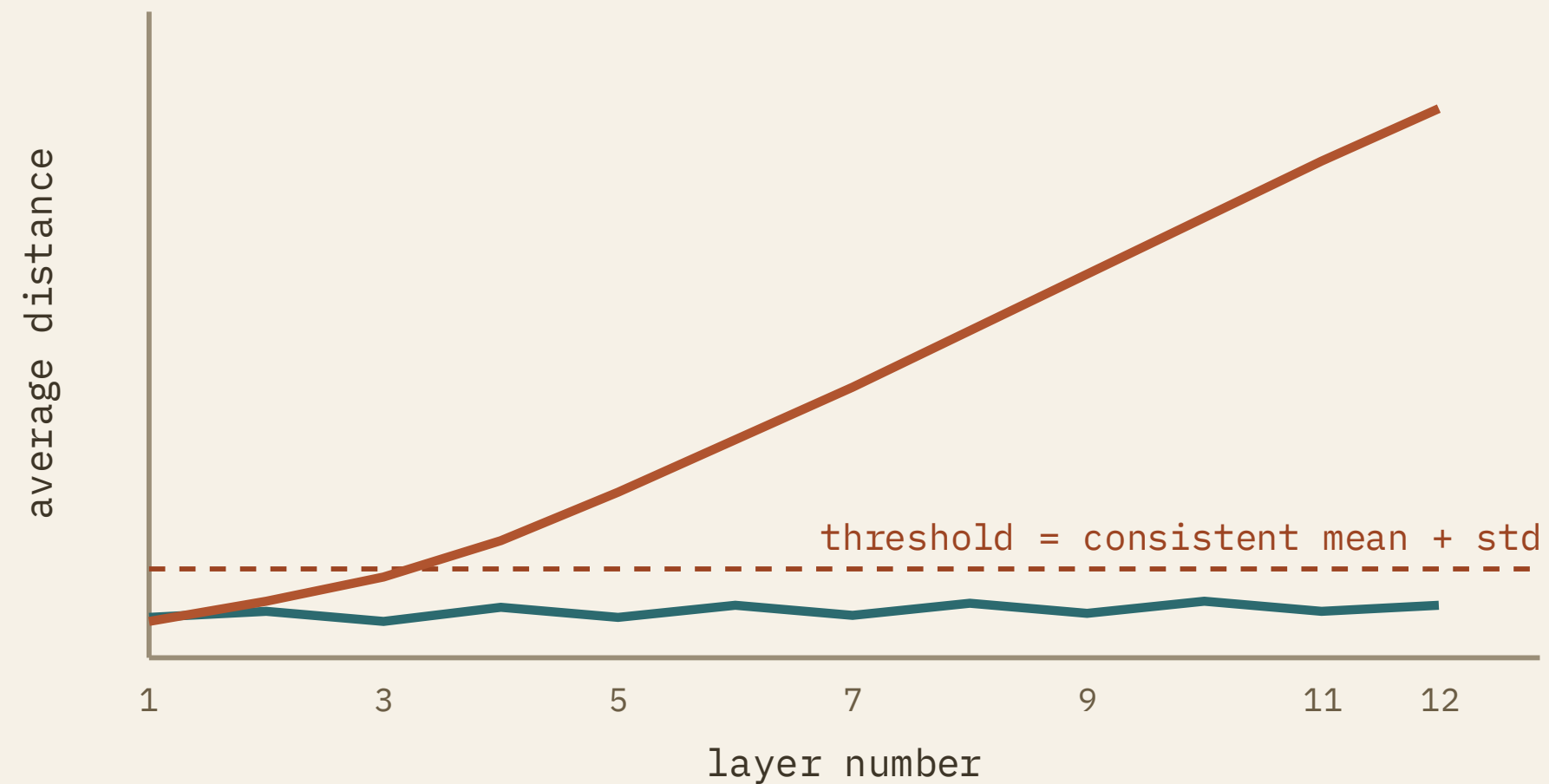
Metric 2 · Layer Distance

At every layer, measure the distance between the hypothesis representation and the paraphrase representation.

If meaning is preserved they stay **close** . We look for the layer where they begin to **drift apart**.



Reading the distance profile



— **Consistent pairs** stay close at every layer.

— **Inconsistent pairs** drift apart as depth grows.

The layer where the orange line crosses the threshold is where the two meanings part ways. That layer is the diagnosis.

What the splitting layer tells us

The depth at which representations separate points to a different stage of the model.

Splits in lower layers

Representation

The wordings are encoded far apart from the very start.

Splits in middle layers

Reasoning

Faulty inference pushes the two meanings apart mid-model.

Splits in upper layers

Classifier

Understood alike, but the final decision differs.

Metric 3 · Credibility, step by step

Papernot & McDaniel, 2018

- 1 From training, store each hypothesis's representation at every layer , a labeled bank.
- 2 For each test paraphrase, at each layer find the K nearest neighbors in that bank.
- 3 Credibility is supporters divided by total neighbors , where total is K times the number of layers.

WORKED EXAMPLE

`12 layers × K=10 = 120 neighbors`

95 of 120 support Entailment → high credibility

What credibility reveals

Compare the paraphrase's credibility to the hypothesis's. The direction of change names the problem.

Credibility drops | Credibility rises wrongly

Generalization → Representation

The paraphrase landed in a region the training data barely covers.

Credibility unchanged

Classifier

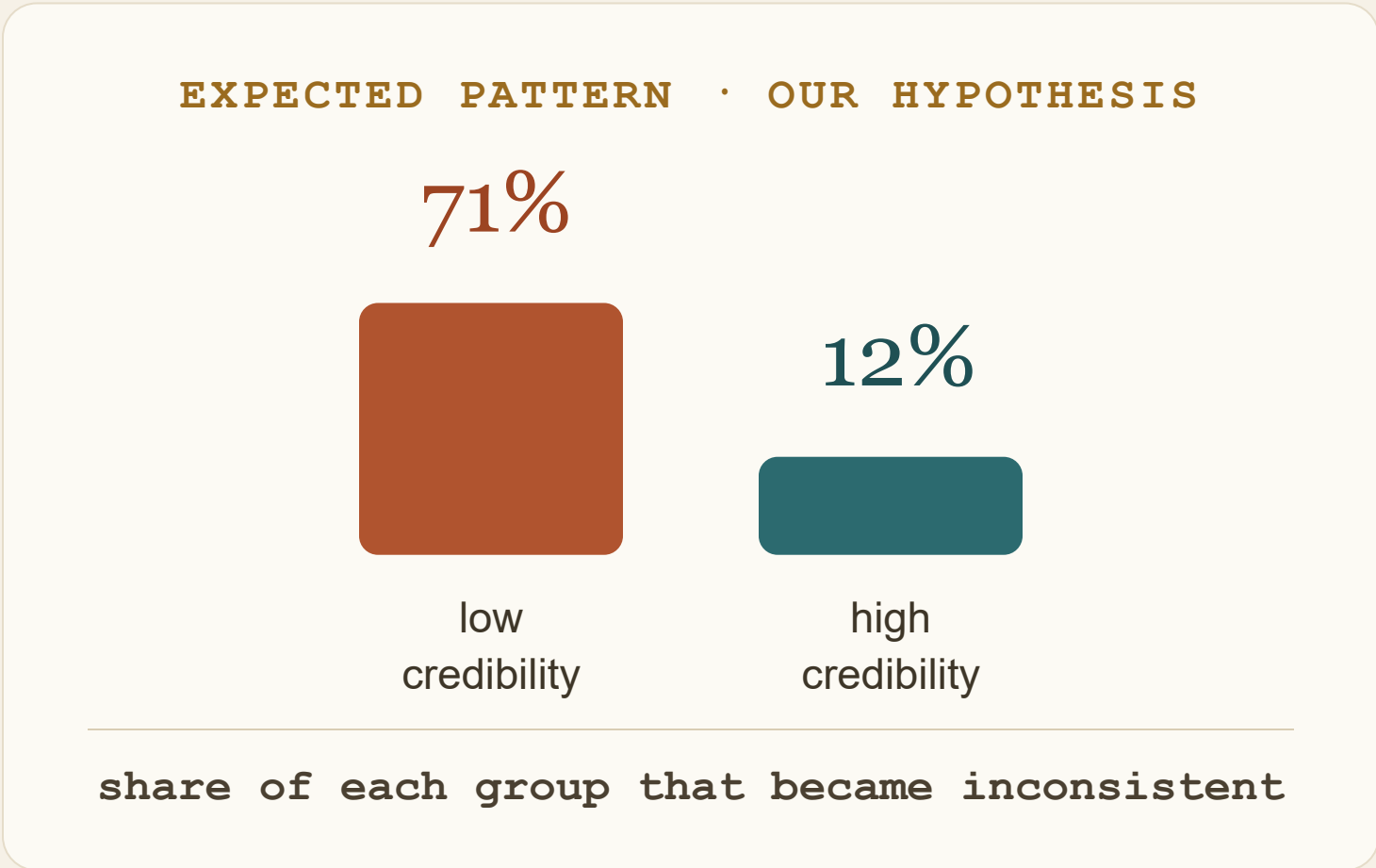
Same internal representation, only the final decision differs.

A change counts as large when it exceeds the typical spread, derived per model and dataset pair.

Phase A • Can we predict it in advance?

Split the inputs into two groups by credibility. In each group separately , what share later became inconsistent under paraphrasing?

The two bars are **separate group rates** , not parts of one whole.
If credibility predicts fragility, the low group is far higher.



Phase B • From diagnosis to a fix

When credibility points to a generalization gap:

- Add targeted paraphrases for the weak hypotheses to the training set.
- Fine-tune , not retrain from scratch, then rebuild the representation bank.

MEASURED AGAINST TWO BASELINES

- 1 The original model, untouched.
- 2 A model given the same number of random paraphrases.

Only if targeted beats random is credibility an informative signal , not just more data.

Two lenses, one diagnosis

Layer Distance

Asks where the representations split, at which layer.

Credibility

Asks where a representation lands, relative to the training data.

They measure different things, so they can disagree, and that is itself a finding. Semantic inconsistency may be *multi-dimensional*, with both representation and classifier contributing. No single cause.

The whole story, in four parts

THE PROBLEM

Strong NLI models flip predictions under meaning-preserving paraphrases, documented widely, but only at the output.

THE GAP

No one has localized where, inside the model , the inconsistency originates.

THE QUESTIONS

Does it begin in the internal representations? Is low credibility the signal? In which layers?

OUR ANGLE OF ATTACK

The first adaptation of Deep k-NN to NLI , diagnose with Layer Distance and Credibility, then a targeted, baseline-controlled fix.

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